

Seasonal Variation in Spatial Extremal Dependence of Wind Speed and Wind Gusts at KNMI Station Locations in the Netherlands

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Abstract

This document formulates a possible master thesis research question on seasonal variation in the spatial extremal dependence of Dutch wind speed and wind gust observations. The central idea is to investigate whether extreme wind events are more spatially widespread in winter than in summer, while explicitly accounting for the changing KNMI station network and possible non-stationarity in the data.

1 Working Title

Seasonal variation in spatial extremal dependence of wind speed and wind gusts at KNMI station locations in the Netherlands.

2 Main Research Question

The main research question is:

Is there seasonal variation in the spatial extremal dependence of wind speed and wind gusts at KNMI station locations in the Netherlands?

An alternative formulation is:

Does the spatial dependence between extreme wind observations at KNMI weather stations differ between seasons, in particular between winter and summer?

3 Interpretation in Plain Language

This thesis investigates whether extreme wind events in the Netherlands tend to be more spatially widespread in some seasons than in others. In practical terms, the question is whether an extreme wind speed or wind gust observed at one KNMI station is more likely to coincide with extreme wind observations at nearby stations during winter than during summer.

The main empirical hypothesis is that winter extremes may be more geographically widespread, while summer extremes may be more localised.

4 Statistical Formulation

Let $X_{s,t}$ denote the wind speed or wind gust observation at station location s and time t . The thesis focuses on the upper tail of the joint distribution of wind observations across stations. The central object of interest is the spatial extremal dependence between locations.

A simplified way to express the relevant conditional extremal dependence is

$$\Pr(X_{j,t} \text{ is extreme} \mid X_{i,t} \text{ is extreme}), \quad (1)$$

where i and j denote two KNMI station locations. The thesis asks whether this conditional extremal dependence changes across seasons and, as a secondary analysis, across longer historical periods.

More generally, the analysis concerns the dependence structure of the upper tail of the multivariate or spatial process

$$\{X_{s,t} : s \in \mathcal{S}, t = 1, \dots, T\}, \quad (2)$$

where $\mathcal{S}$ denotes the set of KNMI station locations included in the analysis.

5 Research Motivation

The Netherlands has a dense network of meteorological stations and a long historical record of hourly wind observations. The available data cover the period from 1951 to 2026 and contain observations of wind speed at 10 metres height and wind gusts. However, the station network is unbalanced: not all current stations were active in the early part of the sample, while by 2026 the network has expanded to approximately 60 stations. This changing spatial network must be taken into account in the empirical design.

Extreme wind events are spatial phenomena. A severe storm may affect a large part of the Netherlands, while other high-wind events may remain more local. This makes spatial extreme value theory a natural framework for analysing the dependence structure of extreme wind speeds and gusts across stations.

The main empirical question is whether spatial extremal dependence is stronger in winter than in summer. In other words, the thesis studies whether winter extremes are more geographically widespread, while summer extremes may be more localised.

6 Connection to Supervisor Feedback

The initial research question was:

Is there temporal variation in the spatial extremal dependence of wind speed at KNMI station locations in the Netherlands?

The supervisor considered this a good starting point, but noted that many theories and methods require stationarity of the data. Therefore, the thesis should explicitly account for possible trends, seasonality, unit roots, and other forms of non-stationarity. One practical way to address this is to focus on specific seasons, such as winter and summer, instead of modelling the entire annual series at once.

The supervisor also suggested sample splitting. This means that the analysis may compare spatial extremal dependence across different periods, for example an early historical period versus a more recent period, while also comparing seasons.

7 Refined Research Question

A refined research question that incorporates this feedback is:

Is there seasonal variation in the spatial extremal dependence of extreme wind speed and wind gusts at KNMI station locations in the Netherlands, and does this dependence change across historical time periods?

This formulation separates two related but distinct questions:

1. **Seasonal variation:** Are extreme wind events more spatially dependent in winter than in summer?
2. **Long-run temporal variation:** Has the strength or range of spatial extremal dependence changed over the period 1951–2026?

8 Possible Sub-questions

The thesis may be organised around the following sub-questions:

1. How strong is the spatial extremal dependence between KNMI stations for extreme wind speed and wind gusts?
2. Does spatial extremal dependence decrease with geographical distance between stations?
3. Is extremal dependence stronger in winter than in summer?
4. Are extreme winter wind events more spatially widespread than extreme summer wind events?
5. Does the estimated spatial dependence structure change when the sample is split into different historical periods?
6. How sensitive are the results to the choice of extreme value method, such as block maxima versus threshold exceedances?
7. How should the changing KNMI station network be handled when estimating spatial extremal dependence over time?

9 Possible Methodological Approach

The thesis can be positioned within spatial extreme value theory. A natural modelling route is to use max-stable processes or related models for spatial extremes. Max-stable processes are useful because they arise as limits for spatial block maxima, analogous to how the generalized extreme value distribution arises for univariate block maxima.

Relevant foundational and methodological references include [Smith \(1990\)](#), [Padoan et al. \(2010\)](#), and [Davison et al. \(2012\)](#). These papers provide a statistical framework for modelling spatial extremes and estimating spatial extremal dependence.

A feasible empirical strategy is:

1. Select a consistent set of KNMI stations or carefully handle the unbalanced station network.
2. Separate the data into seasons, for example winter and summer.
3. Construct extreme observations, for example seasonal or annual block maxima, or threshold exceedances.
4. Estimate spatial extremal dependence separately by season.
5. Compare dependence estimates between winter and summer.
6. Optionally split the sample into historical subperiods to investigate long-run temporal variation.
7. Assess whether the results are robust to station selection, block definition, threshold choice, and time-period splitting.

10 Important Data Issue: Changing Station Network

A key complication is that the KNMI station network changes over time. In 1951, fewer stations were active, while by 2026 the network contains around 60 stations. This means the spatial coverage is not constant over the full sample.

Possible ways to address this include:

1. Use only stations with sufficiently long and reliable records.
2. Perform the main analysis on a balanced subset of stations.
3. Use different balanced station subsets for different time periods.
4. Include a robustness analysis using a larger but unbalanced set of stations.
5. Avoid interpreting changes in dependence as purely climatological if they may partly reflect changes in station coverage.

11 Recommended Final Formulation

The most concise and defensible research question is:

Is spatial extremal dependence in Dutch wind speed and wind gust observations stronger in winter than in summer?

The recommended final research question is:

Is there seasonal variation in the spatial extremal dependence of extreme wind speed and wind gusts at KNMI station locations in the Netherlands?

This version is focused, empirically feasible, and directly incorporates the supervisor's advice to focus on specific seasons. The broader question of long-run temporal change can then be treated as a secondary analysis through sample splitting.

12 Possible Thesis Paragraph

This thesis investigates whether the spatial dependence of extreme wind speed and wind gusts at KNMI station locations in the Netherlands varies across seasons. Using hourly observations from 1951 to 2026, the analysis focuses on the upper tail of the joint distribution of wind observations across a network of meteorological stations. The main question is whether extreme wind events are more spatially widespread in winter than in summer. In statistical terms, the thesis studies whether the probability of observing an extreme wind event at nearby stations, conditional on an extreme event at a given station, differs across seasons. Because the KNMI station network has expanded over time, the empirical analysis must explicitly account for the changing availability of station observations. The methodology will be based on spatial extreme value theory, with particular attention to max-stable processes and likelihood-based inference for spatial extremes.

13 Suggested Thesis Structure

A possible thesis structure is:

1. **Introduction:** Motivation, research question, contribution, and overview.

2. **Data:** KNMI wind speed and wind gust data, station locations, sample period, missingness, and station-network changes.
3. **Methodology:** Spatial extreme value theory, construction of extremes, extremal dependence measures, and max-stable or related models.
4. **Empirical Design:** Seasonal splitting, station selection, block maxima or threshold exceedances, and robustness checks.
5. **Results:** Estimated spatial extremal dependence by season and, optionally, by historical subperiod.
6. **Conclusion:** Summary of findings, limitations, and possible extensions.

References

- Davison, A. C., Padoan, S. A., and Ribatet, M. (2012). Statistical modeling of spatial extremes. *Statistical Science*, 27(2):161–186.
- Padoan, S. A., Ribatet, M., and Sisson, S. A. (2010). Likelihood-based inference for max-stable processes. *Journal of the American Statistical Association*, 105(489):263–277.
- Smith, R. L. (1990). Max-stable processes and spatial extremes. Unpublished manuscript.